SOFTWARE ENGINEER

Brandon Temple

Software engineer focused on automation and data systems, building desktop and web tools with C#, .NET, Python, SQL, JavaScript, and APIs. Experience designing reporting workflows, file-processing utilities, metadata tools, and API-backed internal systems for operational teams.

Starkville, MS btentert@gmail.com brandontemple.com github.com/bt1142msstate

PROFESSIONAL EXPERIENCE

Software Engineer (Data Systems) / Technology Coordinator

MSU Libraries, Mississippi State University | Starkville, MS | Aug 2024 - Present

- Built automation and reporting tools in C#, Python, Perl, SQL, JavaScript, HTML, and CSS to streamline file processing, metadata extraction, data export, and staff workflows for MSU Libraries and Mississippi Library Partnership operations across 12 library systems and 76 branches/campus libraries.
- Developed Query, a browser-based reporting system with configurable filters, API-backed payloads, report history, and Excel export for live library item data.
- Created MetaTable, a WPF metadata scanner for large directory trees with real-time progress tracking and structured Excel exports.
- Translated staff and member-library requests into SirsiDynix API/SQL integrations, documented support workflows, and Excel reports while communicating technical tradeoffs clearly to nontechnical users.
- Applied AI-assisted engineering with C# and C++ systems fundamentals, data structures, debugging, and code review to validate generated code for correctness, performance, and maintainability.

PROJECTS

MetaTable

WPF File Metadata Scanner | C#, .NET 9, EPPlus

WPF scanner for large directory trees; extracts file metadata, tracks progress, and exports structured Excel reports for MSU Libraries.

github.com/bt1142msstate/MetaTable

HTMLConverter

WinUI 3 Batch PDF Converter | .NET 8, WebView2

WinUI 3 app for batch converting HTML files, folders, and ZIP archives to PDF with drag-and-drop intake, scaling controls, and output archiving.

github.com/bt1142msstate/HTMLConverter

Query

Browser-Based Report Builder | JavaScript, HTML, CSS, ExcelJS

Browser-based MSU Libraries reporting tool with configurable filters, structured API payloads, report history, and Excel export.

github.com/bt1142msstate/Query

Ady Resolver

Mississippi Address Resolution Toolkit | Python, Machine Learning

Python address-resolution toolkit and local browser app using cached public data, staged matching, typo-heavy evaluation data, and lightweight ML.

github.com/bt1142msstate/Ady-Resolver

connect4-ai

Browser-Based Game AI Demo | JavaScript, Minimax, Alpha-Beta Pruning

Playable browser game AI with minimax, alpha-beta pruning, replay tooling, experiment modes, and headless validation.

github.com/bt1142msstate/connect4-ai

TECHNICAL SKILLS

Languages	C#, Python, Perl, SQL, JavaScript, TypeScript, HTML, CSS, C++
Frameworks	.NET 8/9, ASP.NET Core, WPF, WinUI 3
Data & APIs	SQL Server, PostgreSQL, EF Core, REST APIs, SirsiDynix APIs, JSON, MARC
Libraries	WebView2, EPPlus, ExcelJS
Tools	Git, Visual Studio, VS Code, Postman, MSTest, ESLint, Playwright
Methods	Data Structures, Algorithms, TDD, Agile, AI-First Engineering, Code Review, Automation, Machine Learning

EDUCATION

Master of Science in Software Engineering
Mississippi State University | Aug 21, 2024 - May 2027 (Expected)

Starkville, MS | GPA: 4.0/4.0

Bachelor of Science in Computer Science, Cum Laude
Mississippi State University | Aug 18, 2021 - May 10, 2024

Starkville, MS | GPA: 3.68/4.0
Honors: Phi Theta Kappa, Gamma Beta Phi

Associate of Arts in Computer Science
East Central Community College | Aug 12, 2019 - May 7, 2021

Decatur, MS | GPA: 3.85/4.0
Honors: Phi Theta Kappa, Gamma Beta Phi
President's List: Fall 2019, Spring 2020
Dean's List: Fall 2020
C++ Programming Award
Outstanding ECCC CS Student Award, awarded by Mississippi State University (custom cowbell)

ADDITIONAL EXPERIENCE

Community Assistant
College View, Mississippi State University | 2022 - 2024

- Supported student residents through communication, policy enforcement, community operations, and incident response.
- Served a 600+ student residential community through crisis response, resident support, and escalation coordination.
- Coordinated resident support workflows while balancing graduate-level software engineering coursework.

STEM Leader & Math Tutor
Afternoon Adventure Learning Center | 2018 - 2021

- Tutored students in math, science, and technology.
- Prepared students for exams and improved study strategies.
- Designed engaging STEM activities with staff.

Computer Science Lab Monitor
East Central Community College | 2019 - 2020

- Maintained and troubleshot lab computers and systems.
- Assisted students with technical issues and software use.
- Supported problem solving and computer literacy.